

Strategic Moves: Example

Consider the following two simultaneous-move strategic form games.

		Player 2				Player 2	
		L	R			L	R
Player 1	U	6, 0	3, 1	Player 1	U	3 , 0	0 , 1
	D	5, 1	2, 0		D	4 , 1	1 , 0
G_1				G_2			

We have $NE(G_1) = \{(U, R)\}$ and $NE(G_2) = \{(D, L)\}$. Assuming that the utility functions represent preferences for outcomes *across* both games, player 1 prefers the NE outcome of G_1 to G_2 even though the utility number for each outcome is lower for G_2 . Now consider the extensive form game Γ in which player 1 first chooses G_1 or G_2 . After observing that choice $i \in \{1, 2\}$, the two players play the simultaneous-move game G_i . Two unambiguous ways to write strategy sets for Γ are

$$S_1 = \{(i, f_1) \mid i \in \{1, 2\}, \text{ and } f_1 : \{1, 2\} \rightarrow \{U, D\}\} \quad \text{and} \quad S_2 = \{f_2 \mid f_2 : \{1, 2\} \rightarrow \{L, R\}\};$$

or

$$S_1 = \{iab \mid i \in \{1, 2\}, a \in \{U, D\}, b \in \{L, R\}\} \quad \text{and} \quad S_2 = \{ab \mid a \in \{L, R\}, b \in \{L, R\}\},$$

where i is the choice of G_i , a is the choice in G_1 and b the choice in G_2 : here $a = f_j(1)$ and $b = f_j(2)$ for $j = 1, 2$.

It follows easily that $SPNE(\Gamma) = \{(2UD, RL)\}$ (or in the first description $SPNE(\Gamma) = \{((2, f_1^*), f_2^*)\}$, where $f_1^*(1) = U$, $f_1^*(2) = D$, $f_2^*(1) = R$, and $f_2^*(2) = L$). The SPNE *outcome* is $h_0 2DL$ (note that people often omit the “ h_0 ” in writing down outcomes).

Clearly player 1 chooses Game 2 in Γ to induce player 2 to choose L in the corresponding subgame. In what sense can we interpret this choice as a strategic move? One benchmark is to consider the *simultaneous-move* game $\tilde{G}$ with strategy sets $S_1 = \{U_1, D_1, U_2, D_2\}$ and $S_2 = \{L, R\}$ with payoffs given by

		Player 2	
		L	R
Player 1	U_1	6, 0	3, 1
	D_1	5, 1	2, 0
	U_2	3, 0	0, 1
	D_2	4, 1	1, 0
$\tilde{G}$			

An interpretation is that player 1 *simultaneously* chooses the game (1 or 2) and an action (U or D) at the same time that player 2 chooses an action (L or R). Since player 2 does not observe 1's choice of a game, that game choice can't influence 2's action. Here 1 has a strictly dominant strategy of U_1 . So $NE(\tilde{G}) = \{(U_1, R)\}$. That player 1 chooses Game 2 in a SPNE of Γ but Game 1 in the unique NE of $\tilde{G}$ leads to the interpretation of 1's choice of Game 2 in Γ as a strategic move.